

Fluid Dynamics of the Atmosphere and Ocean

Problem Set 3

This problem set counts 10% toward your final mark. It is due on 20 February

1. (a) Consider a fluid that obeys the hydrostatic relation

$$\frac{\partial p}{\partial z} = -\rho g.$$

Suppose also that the fluid is an isothermal ideal gas. Show that the density and pressure both diminish exponentially with height. What is the e-folding height? (This is also called the ‘scale height’ of the atmosphere.) Write down an expression for the height, z , as a function of pressure.

- (b) Now suppose that the atmosphere has a uniform lapse rate (i.e., $dT/dz = -\Gamma = \text{constant}$). Show that the height at a pressure p is given by

$$z = \frac{T_0}{\Gamma} \left[1 - \left(\frac{p_0}{p} \right)^{-R\Gamma/g} \right]$$

where T_0 is the temperature at $z = 0$.

- (c) Are the answers you obtained in these two parts consistent in the appropriate limit? Explain.

2. Show that the vertical velocity within a shallow water system is given by

$$w = \frac{z - \eta_b}{h} \frac{Dh}{Dt} + \frac{D\eta_b}{Dt}.$$

Interpret this result, showing that it gives sensible answers at the top and bottom of the fluid layer. If you wish, simply do the problem with $\eta_b = 0$.

3. The formula

$$\left(\frac{d\mathbf{A}}{dt} \right)_I = \left(\frac{d\mathbf{A}}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{A}$$

relates the total rate of change of a vector in an inertial frame (subscript I) to that in a frame rotating with angular velocity $\boldsymbol{\Omega}$ (subscript R). Obtain an expression for

$$\left(\frac{d^2\mathbf{A}}{dt^2} \right)_I$$

in terms of rotating frame quantities. If $\mathbf{A} = \mathbf{r}$, where $\mathbf{r}$ is the position vector, interpret your expression and identify the Coriolis and centrifugal terms.

Hence show that

$$\begin{aligned} \left(\frac{d^3\mathbf{A}}{dt^3} \right)_I &= \left(\frac{d^3\mathbf{A}}{dt^3} \right)_R + 3\boldsymbol{\Omega} \times \left(\frac{d^2\mathbf{A}}{dt^2} \right)_R \\ &\quad + 3\boldsymbol{\Omega} \times \left(\boldsymbol{\Omega} \times \left(\frac{d\mathbf{A}}{dt} \right)_R \right) - |\boldsymbol{\Omega}|^2 \mathbf{A}. \end{aligned}$$

4. In the shallow water equations show that, if the flow is approximately geostrophically balanced, the energy at large scales is predominantly potential energy and that energy at small scales is predominantly kinetic energy. Define precisely what ‘large scale’ and ‘small scale’ mean in this context, and obtain an expression for the transition scale.

5. (a) In an adiabatic shallow water fluid in a rotating reference frame show that the potential vorticity conservation law is

$$\frac{D}{Dt} \frac{\zeta + f}{\eta - h_b} = 0$$

where η is the height of the free surface and h_b is the height of the bottom topography, both referenced to the same flat surface.

- (b) An air column at 60°N with zero relative vorticity ($\zeta = 0$) stretches from the surface to the tropopause, which we assume is a rigid lid, at 10 km. The air column moves zonally on to a plateau 2.5 km high. What is its relative vorticity? Suppose it then moves southwards to 30°N . What is its vorticity? (Assume that the density is constant.)

6. The shallow water equations, linearized about a state of rest, may be written as

$$\begin{aligned} \frac{\partial u'}{\partial t} - f_0 v' &= -g \frac{\partial \eta'}{\partial x}, & \frac{\partial v'}{\partial t} + f_0 u' &= -g \frac{\partial \eta'}{\partial y}, \\ \frac{\partial \eta'}{\partial t} + H \left(\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} \right) &= 0. \end{aligned}$$

Suppose there is a solid boundary (e.g., a coastline) at $y = 0$, and look for solutions that have $u' = 0$ everywhere, and with $f_0 < 0$. Show that the resulting waves are non-dispersive and travel at a speed $c = \sqrt{gH}$. Is the coastline to the left or to the right of the direction of travel? Suppose that these waves are generated just off the shore of Perth, Australia. Do they move north or south?

7. In a self-gravitating spherical fluid, like a star, hydrostatic balance may be written

$$\frac{\partial p}{\partial r} = -\frac{GM(r)}{r^2} \rho,$$

where $M(r)$ is the mass interior to a sphere of radius r , and G is a constant. Obtain an expression for the pressure as a function of radius when the fluid has constant density. [Optional: obtain the pressure as a function of radius if the fluid is an isothermal ideal gas, if possible. Discuss your solution, or your failure to find one.]